

A Parameter-Free Multi-View Clustering Framework With Adaptive Anchors for Large-Scale Data

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Abstract—Anchor-based multi-view clustering has gained increasing attention for its efficiency in approximating similarity structures and scaling to large datasets. To reduce the burden of manual hyper-parameter tuning, recent studies have introduced parameter-free extensions. However, existing methods still face critical challenges: anchors are typically fixed after initialization, limiting adaptability to heterogeneous data; enforcing a shared anchor set across views suppresses view-specific diversity; and heuristic or self-weighted fusion strategies often lack explicit cross-view alignment, resulting in structural inconsistencies. To address these issues, we propose a Parameter-Free Multi-view Clustering framework with Adaptive Anchors for Large-scale Data (FPMCAA). Unlike existing approaches that decouple anchor construction and graph fusion, FPMCAA integrates adaptive anchor learning, anchor graph construction, and explicit cross-view alignment within a unified optimization model. Anchors are iteratively refined to capture complex distributions, while view-specific graphs are aligned toward a consensus structure without sacrificing inherent diversity. The framework avoids manual hyperparameter tuning and achieves linear computational complexity through efficient alternating optimization. Extensive experiments on benchmark datasets demonstrate that FPMCAA consistently outperforms state-of-the-art methods in clustering performance, robustness, and scalability.

Index Terms—Multi-view clustering, adaptive anchor graphs, graph alignment, parameter-free learning, large-scale data.

I. INTRODUCTION

WITH the rapid development of data collection and sensing technologies, information is increasingly gathered from multiple heterogeneous sources and represented through

diverse feature modalities, also referred to as views [1], [2], [3], [4]. For example, an image can be simultaneously described by color histograms, texture features, and shape descriptors, while news may appear in multiple languages, such as English, Chinese, and French. Traditional single-view clustering methods rely on a single feature representation to partition samples, which often fails to uncover the intrinsic structure of data and leads to significant performance degradation under noise interference or missing data. In contrast, multi-view clustering (MVC) has attracted considerable attention in recent years because of its ability to leverage both the consistency and complementarity across multiple views, thereby achieving superior clustering performance [1], [5], [6], [7], [8]. It has found applications in a wide range of domains, including computer vision, natural language processing, bioinformatics, and social network analysis [9], [10], [11], [12], [13], [14].

To exploit multi-view information, researchers have developed a large variety of MVC approaches. Subspace-based methods aim to project multi-view data into shared or view-specific low-dimensional subspace representations, thereby capturing latent structures and improving robustness against noise and discrepancies across views [15], [16], [17], [18], [19]. For instance, Lan et al. [19] pursued a consistent low-rank subspace across views via trifactor coefficient decomposition. Graph-based methods emphasize modeling the local geometric structure of data and maintaining consistency across views [20], [21], [22], [23], [24]. Tang et al. [20] learn view-wise affinity graphs and unify graph construction with partition generation via spectral rotation in one step. Tensor-based methods go beyond pairwise relationships by modeling high-order correlations among views [25], [26], [27], [28], [29], [30]. For example, Qin et al. [30] simultaneously encode pairwise and high-order inter-view correlations by stacking similarity matrices into a single tensor and constraining it with a low-rank tensor norm. While these approaches achieve promising results, they generally involve high computational costs, typically $O(n^2)$ or even $O(n^3)$, along with complex optimization, which limit their scalability to large-scale datasets. To improve efficiency, researchers have turned their attention to scalable frameworks, among which anchor-based methods have emerged as a representative direction.

While the aforementioned approaches have achieved impressive clustering results, anchor-based MVC has attracted particular interest for its scalability and efficiency on large-scale datasets [31], [32], [33], [34], [35]. The key idea is to introduce

Received 7 October 2025; revised 10 March 2026; accepted 2 June 2026. Date of publication 5 June 2026; date of current version 8 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62572199, Grant 92467109, Grant 62476101, and Grant U21A20478, in part by the Major Key Project of PCL under Grant PCL2025A11 and Grant PCL2025A13, in part by the National Key R&D Program of China under Grant 2023YFA1011601, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2024A1515140137. Recommended for acceptance by F. Banaei-Kashani. (Corresponding author: Zhiwen Yu.)

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The source code of FPMCAA is available at <https://github.com/Xuchen2020/FPMCAA>.

Digital Object Identifier 10.1109/TKDE.2026.3700824

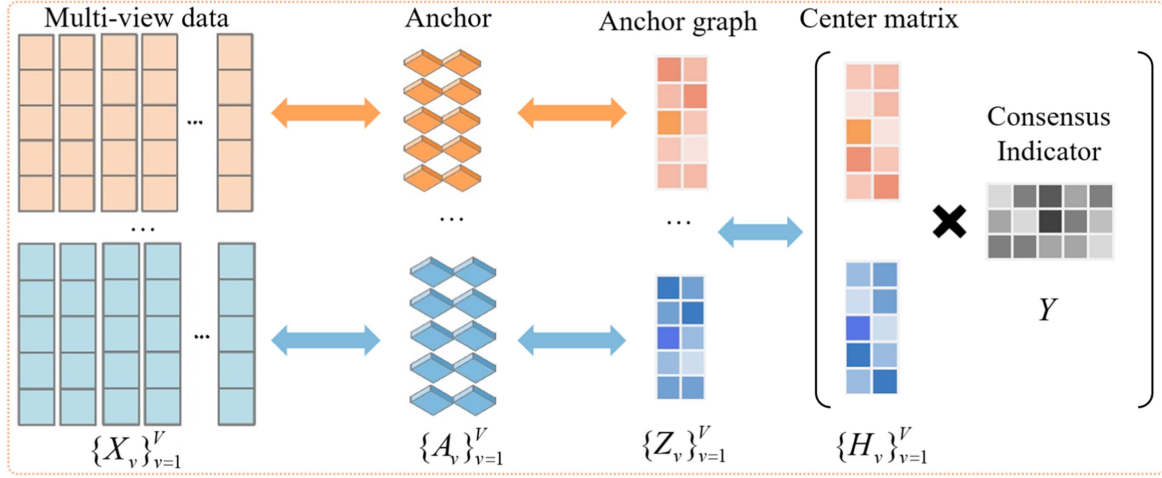


Fig. 1. Illustration of the proposed FPMCAA framework. Given multi-view data X_v , view-specific anchors A_v are first constructed, from which anchor graphs Z_v are learned. The anchor graphs are then factorized into view-specific center matrices H_v and a consensus indicator matrix Y to achieve cross-view alignment. All variables are jointly optimized in a unified framework, enabling them to mutually refine each other throughout the learning process.

a small set of representative data points, known as anchors, to approximate the overall data structure. This strategy dramatically simplifies graph construction while preserving essential structural patterns. For example, Kang et al. proposed the Large-scale Multi-View Subspace Clustering (LMVSC) [34], which constructs view-specific anchor graphs using k -means centers as anchors, achieving linear-time scalability. Qin et al. later introduced Dual Consensus Anchor Learning (DALF) [35], where a shared anchor set is learned across all views to enforce clustering structure correspondence.

Nevertheless, most existing anchor-based methods still rely on manually tuned hyper-parameters (e.g., trade-off coefficients or regularization parameters), which not only complicates model selection but also limits their practical applicability. To address this issue, researchers have recently explored parameter-free anchor-based frameworks. For instance, NpGC introduces two types of anchors (view-related and view-unrelated) and learns a consensus bipartite graph to capture both complementary and consistent information [36]. FPMVS constructs a common anchor set shared across all views and directly learns a consistency-preserving anchor graph for large-scale clustering [37]. MvSST adopts a fixed-anchor bipartite strategy, first building discrete bipartite graphs for each view and then fusing them via a self-weighted mechanism [38]. These studies significantly enhance the practicality of anchor-based MVC, although challenges such as adaptability to heterogeneous views and robustness to noise remain open for further exploration.

Despite these advances, existing methods still face several challenges. First, anchors are typically fixed after initialization, which limits their adaptability to heterogeneous data distributions. Second, forcing all views to share a single set of common anchors and anchor graphs often ignores the inherent diversity among views, thereby weakening the ability to capture complementary information. Finally, although heuristic or self-weighted fusion strategies can be effective to some extent, they typically lack explicit alignment across

anchors of different views, leaving potential inconsistencies unresolved.

Motivated by these challenges, this paper proposes a parameter-free multi-view clustering framework with adaptive anchors for large-scale data, termed FPMCAA and illustrated in Fig. 1. Unlike prior approaches that decouple anchor construction, anchor graph learning, and graph fusion into independent stages, FPMCAA integrates adaptive anchor learning, anchor graph construction, and explicit cross-view alignment into a unified optimization framework, enabling synergistic interaction among all components during iteration. Specifically, FPMCAA constructs view-specific anchors and learns corresponding anchor graphs for each view, thereby preserving the intrinsic geometric structure and local data distribution of individual modalities. Meanwhile, a consistency-driven alignment constraint is imposed on the view-specific anchor graphs to progressively guide them toward a shared global consensus structure. The complementary structural information uncovered during alignment is further leveraged to refine the anchor graphs, enhancing the quality and expressiveness of the unified representation. In addition, a Frobenius norm regularization term is incorporated into the anchor graphs to suppress degenerate solutions and enforce structural compactness and discriminability, thereby improving robustness under noise and inter-view heterogeneity. Compared with conventional anchor-based methods that rely on manually tuned hyper-parameters or heuristic fusion schemes, FPMCAA achieves an automatic balance between modeling flexibility and alignment constraints without external parameter intervention. Owing to this unified and self-adaptive design, the proposed framework demonstrates superior clustering performance, robustness, and scalability in large-scale and heterogeneous multi-view scenarios.

Our main contributions are summarized below:

- We propose a parameter-free multi-view clustering framework that unifies adaptive anchor learning, anchor graph construction, and explicit cross-view alignment within a

TABLE I
SUMMARY OF NOTATION

Symbol	Descriptions
n, V, l, k	The number of samples, views, anchors, and clusters
d_v	Feature dimension of the v -th view
$\alpha \in \mathbb{R}^{V \times 1}$	View weight vector
$\mathbf{X}_v \in \mathbb{R}^{d_v \times n}$	Data matrix in the v -th view
$\mathbf{A}_v \in \mathbb{R}^{d_v \times l}$	The anchor matrix for the v -th view
$\mathbf{Z}_v \in \mathbb{R}^{l \times n}$	The anchor graph matrix for the v -th view
$\mathbf{H}_v \in \mathbb{R}^{l \times k}$	The projection matrix for the v -th view
$\mathbf{Y} \in \mathbb{R}^{k \times n}$	The latent cluster indicator

single optimization model, enabling cooperative enhancement among modules.

- The proposed FPMCAA framework separately learns anchors and anchor graphs for each view to preserve their intrinsic geometric structures, and progressively drives them toward a consensus representation through a graph alignment strategy. Meanwhile, complementary information uncovered during alignment is fed to refine the anchor graphs, thereby jointly capturing both consistency and complementarity across views.
- We conduct theoretical analyses on convergence and computational complexity to ensure the soundness of the proposed framework. Comprehensive experiments on several benchmark datasets further verify its superiority in clustering accuracy, performance, and scalability.

The remainder of this paper is organized as follows. Section II briefly reviews related work, focusing on multi-view anchor clustering. Section III presents FPMCAA methodology, covering model formulation, optimization, complexity, and convergence analysis. Section IV provides extensive experiments on nine datasets, including experimental settings, clustering performance comparison, sensitivity to the number of anchors, ablation study, running-time comparison, and convergence verification. Section V concludes the paper and discusses future research directions.

II. RELATED WORKS

A. Notations

For convenience, the main mathematical notations used in this paper are summarized in Table I.

B. Multi-View Clustering

To exploit complementary information across multiple views, a large body of research has been devoted to multi-view clustering (MVC). Among them, subspace-based approaches constitute one of the most representative frameworks. These methods assume that multi-view data lie in a union of low-dimensional subspaces and learn a self-representation matrix to capture the latent relationships among samples. Given multi-view data $\mathbf{X}$ consisting of V distinct views, where $\mathbf{X}_v \in \mathbb{R}^{d_v \times n}$ denotes the data matrix of the v -th view with d_v features and n samples, a typical formulation can be written as

$$\min_{\mathbf{S}_v} \sum_{v=1}^V \|\mathbf{X}_v - \mathbf{X}_v \mathbf{S}_v\|_F^2 + \lambda \Omega(\mathbf{S}_v) \quad (1)$$

where $\mathbf{S}_v \in \mathbb{R}^{n \times n}$ denotes the self-representation matrix and $\Omega(\cdot)$ is a regularization term encouraging desirable structures such as sparsity or low-rankness. Based on this formulation, numerous variants have been developed to capture shared and complementary structures across views, such as low-rank subspace learning [15], latent representation learning [16], and collaborative subspace modeling [17], [18].

Closely related to subspace-based methods, graph-based MVC approaches focus on constructing similarity graphs to preserve the local geometric structure of data in each view. These methods typically learn view-wise affinity matrices and integrate them to obtain a consensus representation or clustering structure [20], [21], [22], [24]. By explicitly modeling neighborhood relationships, graph-based methods are effective in capturing manifold structures and improving clustering robustness. Despite their promising performance, most subspace- and graph-based MVC methods rely on constructing or manipulating $n \times n$ similarity matrices, leading to high computational and memory costs for large-scale data. To address this limitation, anchor-based clustering methods have been introduced to improve scalability for large datasets.

Anchor-based MVC methods approximate the full similarity graph with a small set of representative anchors, significantly reducing computational complexity while preserving essential structural information. The flexible anchor graph can be constructed by minimizing the following objective function

$$\begin{aligned} \min_{\mathbf{Z}_v} \sum_{v=1}^V \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \lambda \Omega(\mathbf{Z}_v), \\ \text{s.t. } \mathbf{Z}_v \geq 0, \mathbf{Z}_v \mathbf{1} = \mathbf{1}, \end{aligned} \quad (2)$$

where $\mathbf{A}_v \in \mathbb{R}^{d_v \times l}$ is the anchor matrix containing l anchors, and $\mathbf{Z}_v \in \mathbb{R}^{l \times n}$ is the anchor graph.

Within this paradigm, a number of representative methods have been proposed. Early studies adopt fixed anchor selection strategies, using k-means centroids [34], [39] or direct alternating sampling [40] to obtain anchors. However, since anchors remain fixed after initialization, these methods lack adaptability to heterogeneous view distributions and may lead to suboptimal representations. To mitigate the limitations of view-specific fixed anchors and further enhance cross-view consistency, several studies have attempted to construct a shared anchor set across all views [41], [42], where the anchors are typically updated during optimization rather than remaining fixed. By enforcing a common anchor basis, these methods seek to align heterogeneous feature spaces at the graph construction stage, thereby facilitating the discovery of more unified clustering structures. For example, Chen et al. [41] jointly learn anchors, graphs, and clustering assignments within a unified framework, while Ji et al. [42] employ shared anchors but learn view-specific anchor graphs and further exploit tensor regularization to capture higher-order cross-view correlations. Although shared-anchor strategies improve alignment, enforcing all views to rely on the same anchors may suppress view-specific geometric structures and limit the capacity to capture complementary information.

To better maintain view-specific characteristics while retaining scalability, recent methods have explored adaptive and view-specific anchor construction. For instance, Liu et al. [43] propose a flexible anchor graph framework that decomposes each anchor graph into consistent and view-private components, while Wang et al. [44] design a flexible anchor graph fusion model that aligns view-specific graphs through structural regularization. While enhancing flexibility, they often introduce additional parameters for cross-view alignment, increasing model complexity. To alleviate this issue, several recent studies have attempted to design parameter-free anchor-based models. For example, MvSST constructs fixed bipartite graphs for each view and performs graph fusion through a self-weighted strategy without introducing external tuning parameters. FPMVS and EOMSC [45] directly build a single consensus anchor graph from shared anchors, thereby avoiding graph-level alignment or fusion weights. NpGC further decomposes the bipartite graph into correlated and uncorrelated components without introducing manually tuned trade-off parameters.

Nevertheless, most existing methods still treat anchor construction, graph learning, and cross-view fusion as relatively independent stages, which may limit the interaction among these components during optimization. In contrast, the proposed framework integrates these components into a unified parameter-free learning framework, enabling anchors, anchor graphs, and the consensus representation to be jointly refined during optimization.

III. PROPOSED METHODOLOGY

This section presents the proposed parameter-free adaptive anchor framework, termed FPMCAA. The overall methodology is organized into three components. First, we formulate a unified objective that simultaneously integrates adaptive anchor learning, anchor graph construction, and explicit cross-view alignment. Then, an alternating optimization strategy with closed-form updates is developed to efficiently solve the model. Finally, we provide theoretical analysis on the computational complexity and convergence properties.

A. The FPMCAA Model

Multi-view data typically exhibit discrepancies in feature dimensionality and statistical properties, rendering their effective utilization a non-trivial challenge. Many anchor-based methods adopt a shared or predefined anchor set for convenience; this practice tends to overlook view-specific structural patterns and incurs a partial loss of structural information. Additionally, some graph-alignment constraints inevitably introduce at least one hyper-parameter, resulting in increased model complexity.

To address these issues in an integrated way, we design a parameter-free multi-view clustering framework (FPMCAA) that unifies view-specific anchor learning, anchor-graph construction, and cross-view consensus alignment within a single optimization model.

For each view v , we simultaneously learn the anchor bases $\mathbf{A}_v \in \mathbb{R}^{d_v \times l}$ and the corresponding anchor graph $\mathbf{Z}_v \in \mathbb{R}^{l \times n}$ by minimizing the reconstruction error while encouraging compact

assignments

$$\begin{aligned} \min_{\alpha, \mathbf{A}_v, \mathbf{Z}_v} \quad & \sum_{v=1}^V \alpha_v^2 \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v\|_F^2 \\ \text{s.t.} \quad & \alpha^T \mathbf{1} = 1, \alpha \geq 0, \mathbf{Z}_v^T \mathbf{1} = 1, \mathbf{Z}_v \geq 0, \\ & \mathbf{A}_v^T \mathbf{A}_v = \mathbf{I}. \end{aligned} \quad (3)$$

where $\alpha \in \mathbb{R}^{V \times 1}$ is the coefficient that adjusts the contribution of each view. The α_v^2 amplifies the contribution of views with lower reconstruction error, which typically exhibit clearer structural patterns beneficial for clustering, while suppressing noisy views. The Frobenius regularizer on $\mathbf{Z}_v$ prevents coefficient explosion and promotes stable, low-redundancy anchors, while the constraint $\mathbf{Z}_v^T \mathbf{1} = \mathbf{1}$ guarantees that each sample is represented by a normalized combination of anchors, enabling a soft-assignment interpretation. The orthogonality constraint on $\mathbf{A}_v$ avoids degenerate bases and preserves interpretability, making the learned anchors more discriminative and diverse.

Although view-specific anchor graphs capture local geometry faithfully, they may encode inconsistent or even conflicting cluster structures. Unlike fusion strategies that combine view-specific graphs through fixed weights or heuristic consensus rules, we introduce a common latent cluster indicator $\mathbf{Y} \in \mathbb{R}^{k \times n}$, where k denotes the number of clusters, along with view-specific projection matrices $\mathbf{H}_v \in \mathbb{R}^{l \times k}$ and explicitly align each anchor graph to a latent consensus space

$$\begin{aligned} \min_{\mathbf{H}_v, \mathbf{Y}} \quad & \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2 \\ \text{s.t.} \quad & \mathbf{Y}^T \mathbf{Y} = \mathbf{I}. \end{aligned} \quad (4)$$

This formulation enables each anchor graph to be aligned to a common latent space through joint optimization rather than relying on explicit graph combination strategies. The orthogonality of $\mathbf{Y}$ preserves compactness and avoids redundancy in the latent representation. In this way, cross-view information is fused implicitly through the alignment constraint, which encourages view-specific anchor graphs to be consistent with a shared latent cluster structure while still preserving complementary information across views.

Combining the above components, the final unified objective of FPMCAA is

$$\begin{aligned} \min_{\alpha, \mathbf{A}_v, \mathbf{Z}_v, \mathbf{H}_v, \mathbf{Y}} \quad & \sum_{v=1}^V \alpha_v^2 \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v\|_F^2 \\ & + \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2 \\ \text{s.t.} \quad & \alpha^T \mathbf{1} = 1, \alpha \geq 0, \mathbf{Z}_v^T \mathbf{1} = 1, \mathbf{Z}_v \geq 0, \\ & \mathbf{A}_v^T \mathbf{A}_v = \mathbf{I}, \mathbf{Y}^T \mathbf{Y} = \mathbf{I}. \end{aligned} \quad (5)$$

The balance among these terms is implicitly regulated by the adaptive view weights α together with the normalization and orthogonality constraints imposed on the variables, which help maintain comparable scales among different components of the objective and thus avoid introducing additional trade-off parameters. Here, the cluster number k is treated as a task-dependent input, as in most clustering formulations, rather than a balancing

coefficient within the objective function. Benefiting from the anchor-based linear complexity, FPMCAA scales gracefully to large-scale datasets while maintaining high clustering performance and interpretability.

B. Optimization

The objective function in (5) is non-convex when considering all variables jointly. However, it is convex with respect to each individual variable when the others are fixed. Therefore, we adopt an alternating optimization strategy, updating each variable iteratively to guarantee the convergence of the overall objective.

Update $\mathbf{A}_v$: With other variables fixed, optimizing $\mathbf{A}_v$ is simplified to solving the following problem

$$\begin{aligned} \min_{\mathbf{A}_v} \quad & \sum_{v=1}^V \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 = \max_{\mathbf{A}_v} \text{Tr}(\mathbf{A}_v^T \mathbf{X}_v \mathbf{Z}_v^T) \\ \text{s.t.} \quad & \mathbf{A}_v^T \mathbf{A}_v = \mathbf{I}. \end{aligned} \quad (6)$$

The optimal solution is obtained by Singular Value Decomposition (SVD): Let $\mathbf{X}_v \mathbf{Z}_v^T = \mathbf{U}_G \mathbf{\Sigma}_G \mathbf{V}_G^T$, then

$$\mathbf{A}_v = \mathbf{U}_G \mathbf{V}_G^T. \quad (7)$$

Update $\mathbf{Z}_v$: Fixing the other variables, the subproblem for $\mathbf{Z}_v$ is

$$\begin{aligned} \min_{\mathbf{Z}_v} \quad & \alpha_v^2 \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2 \\ \text{s.t.} \quad & \mathbf{Z}_v \geq 0, \mathbf{Z}_v^T \mathbf{1} = \mathbf{1}. \end{aligned} \quad (8)$$

Ignoring the constraints and setting the derivative to zero, this gives an unconstrained minimizer

$$\mathbf{Z}_v = (\alpha_v^2 \mathbf{A}_v^T \mathbf{A}_v + 2\mathbf{I})^{-1} (\alpha_v^2 \mathbf{A}_v^T \mathbf{X}_v + \mathbf{H}_v \mathbf{Y}). \quad (9)$$

Since $\mathbf{A}_v^T \mathbf{A}_v = \mathbf{I}$, this reduces to

$$\mathbf{U}_v = (\alpha_v^2 + 2)^{-1} (\alpha_v^2 \mathbf{A}_v^T \mathbf{X}_v + \mathbf{H}_v \mathbf{Y}). \quad (10)$$

To satisfy the constraints in (8), each column $\mathbf{u}_v^{(j)}$ of $\mathbf{U}_v$ is projected onto the probability simplex

$$\Delta = \{\mathbf{z} \in \mathbb{R}^l \mid \mathbf{z} \geq 0, \mathbf{1}^T \mathbf{z} = 1\}. \quad (11)$$

Equivalently, the constrained solution under the Karush-Kuhn-Tucker (KKT) conditions can be written as

$$\mathbf{z}_v^{(j)} = \text{Proj}_{\Delta}(\mathbf{u}_v^{(j)}) = \max(\mathbf{u}_v^{(j)} - \tau_j \mathbf{1}, 0), \quad (12)$$

where the scalar τ_j is the unique value satisfying

$$\sum_{i=1}^l \max(u_{v,i}^{(j)} - \tau_j, 0) = 1, \quad (13)$$

and $\Delta = \{\mathbf{z} \in \mathbb{R}^l \mid \mathbf{z} \geq 0, \mathbf{1}^T \mathbf{z} = 1\}$.

Where τ_j is the unique scalar that enforces the sum-to-one constraint. In practice, τ_j can be obtained efficiently by the sorting-based simplex projection algorithm [46]. Repeating this for all columns yields the updated $\mathbf{Z}_v$.

Proposition 1: Let $\mathbf{U}_v$ be defined in (10). The solution of (8) is obtained columnwise by

$$\mathbf{z}_v^{(j)} = \text{Proj}_{\Delta}(\mathbf{u}_v^{(j)}) = \max(\mathbf{u}_v^{(j)} - \tau_j \mathbf{1}, 0), \quad (14)$$

where $\mathbf{u}_v^{(j)}$ denotes the j -th column of $\mathbf{U}_v$, $\Delta = \{\mathbf{z} \in \mathbb{R}^l \mid \mathbf{z} \geq 0, \mathbf{1}^T \mathbf{z} = 1\}$, and τ_j is the unique scalar satisfying

$$\sum_{i=1}^l \max(u_{v,i}^{(j)} - \tau_j, 0) = 1. \quad (15)$$

Proof: Define the objective (with other variables fixed)

$$\begin{aligned} \min_{\mathbf{Z}_v} \quad & \alpha_v^2 \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2 \\ \text{s.t.} \quad & \mathbf{Z}_v \geq 0, \mathbf{Z}_v^T \mathbf{1} = \mathbf{1}. \end{aligned} \quad (16)$$

Using $\mathbf{A}_v^T \mathbf{A}_v = \mathbf{I}$ and expanding only the terms depending on $\mathbf{Z}_v$, we obtain

$$\begin{aligned} f(\mathbf{Z}_v) = \text{constant} + (\alpha_v^2 + 2) \|\mathbf{Z}_v\|_F^2 \\ - 2 \langle \mathbf{Z}_v, \alpha_v^2 \mathbf{A}_v^T \mathbf{X}_v + \mathbf{H}_v \mathbf{Y} \rangle. \end{aligned} \quad (17)$$

Let $\mathbf{B}_v := \alpha_v^2 \mathbf{A}_v^T \mathbf{X}_v + \mathbf{H}_v \mathbf{Y}$ and define $\mathbf{U}_v$ as in (10). Completing the square yields

$$f(\mathbf{Z}_v) = \text{constant} + (\alpha_v^2 + 2) \|\mathbf{Z}_v - \mathbf{U}_v\|_F^2. \quad (18)$$

Thus minimizing $f(\mathbf{Z}_v)$ subject to $\mathbf{Z}_v \geq 0$ and $\mathbf{Z}_v^T \mathbf{1} = \mathbf{1}$ is equivalent to

$$\min_{\mathbf{Z}_v} \|\mathbf{Z}_v - \mathbf{U}_v\|_F^2 \quad \text{s.t.} \quad \mathbf{Z}_v \geq 0, \mathbf{Z}_v^T \mathbf{1} = \mathbf{1}. \quad (19)$$

Because the Frobenius norm separates across columns, the problem decouples columnwise: for each column j ,

$$\min_{\mathbf{z} \in \mathbb{R}^l} \|\mathbf{z} - \mathbf{u}_v^{(j)}\|_2^2 \quad \text{s.t.} \quad \mathbf{z} \geq 0, \mathbf{1}^T \mathbf{z} = 1, \quad (20)$$

which is the Euclidean projection of $\mathbf{u}_v^{(j)}$ onto the probability simplex Δ . The KKT conditions for this constrained quadratic imply the componentwise form

$$z_i = \max(u_{v,i}^{(j)} - \tau_j, 0), \quad (21)$$

together with (15) that enforces the sum-to-one constraint. The scalar τ_j exists and is unique (see the monotonicity of $\tau \mapsto \sum_i \max(u_i - \tau, 0)$); an efficient $O(l \log l)$ sorting-based algorithm to compute τ_j is given in Duchi et al. (2008). Therefore (14) holds for every column, which completes the proof. $\square$

Update $\mathbf{H}_v$: Fixing the other variables, the subproblem for optimizing $\mathbf{H}_v$ becomes:

$$\min_{\mathbf{H}_v} \sum_{v=1}^V \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2. \quad (22)$$

Setting the derivative of (22) to zero, the closed-form solution is given by

$$\mathbf{H}_v = \mathbf{Z}_v \mathbf{Y}^T. \quad (23)$$

Algorithm 1: Optimization Algorithm of FPMCAA.

Input: the multi-view dataset $\{\mathbf{X}^1, \mathbf{X}^2, \dots, \mathbf{X}^V\}$, cluster number k .

- 1: Initial: $\mathbf{Z}_v, \mathbf{A}_v, \mathbf{H}_v, \mathbf{Y}, \alpha$.
- 2: **while** not converged **do**
- 3: update $\mathbf{A}_v$ by (7);
- 4: update $\mathbf{Z}_v$ by (12);
- 5: update $\mathbf{H}_v$ by (23);
- 6: update $\mathbf{Y}$ by (26);
- 7: update α by (28);
- 8: **end while**

Output: Performing k-means on $\mathbf{Y}$ to obtain final clustering results.

Update $\mathbf{Y}$: Fixing the other variables, the optimization problem for $\mathbf{Y}$ is formulated as:

$$\min_{\mathbf{Y}} \sum_{v=1}^V \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2 \quad \text{s.t.} \quad \mathbf{Y}^T \mathbf{Y} = \mathbf{I}. \quad (24)$$

Similar to the optimization of $\mathbf{A}_v$, This can also be rewritten as the following equivalent trace problem

$$\max_{\mathbf{Y}} \text{Tr}(\mathbf{Y}^T \mathbf{M}) \quad \text{s.t.} \quad \mathbf{Y}^T \mathbf{Y} = \mathbf{I}. \quad (25)$$

where $\mathbf{M} = \sum_{v=1}^V \mathbf{H}_v^T \mathbf{Z}_v$.

The optimal solution is obtained via SVD: Let $\mathbf{M} = \mathbf{U} \Sigma \mathbf{V}^T$, then the closed-form solution is:

$$\mathbf{Y} = \mathbf{U} \mathbf{V}^T. \quad (26)$$

Update α : Fixing the other variables, the optimization problem for α is formulated as

$$\min_{\alpha} \sum_{v=1}^V \alpha_v^2 c_v \quad \text{s.t.} \quad \alpha^T \mathbf{1} = 1, \alpha \geq 0. \quad (27)$$

where $c_v = \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2$. This constrained quadratic optimization problem admits a closed-form solution under the standard simplex condition. The reconstruction error c_v quantifies the approximation fidelity of the v -th view, thereby enabling an adaptive weighting mechanism in which views with lower reconstruction residuals are assigned greater importance during the joint optimization process. The optimal closed-form solution is given by

$$\alpha_v = \frac{\frac{1}{c_v}}{\sum_{v=1}^V \frac{1}{c_v}} \quad (28)$$

To facilitate implementation and understanding, the optimization process of FPMCAA model is formally described in Algorithm 1. Owing to the closed-form nature of every update step, the optimization dispenses with nested iterative solvers, stabilises convergence, and remains free of additional hyper-parameters, thereby enhancing the practical deployability of the proposed framework.

C. Complexity Analysis

1) *Time Complexity*: In each iteration, the main computational cost comes from updating the anchor graphs $\mathbf{Z}_v$, anchor bases $\mathbf{A}_v$, alignment matrices $\mathbf{H}_v$, the consensus indicator $\mathbf{Y}$, and the weight vector α . For the v -th view, computing $\mathbf{U}_v$ and projecting it onto the simplex requires matrix multiplications of order $O(d_v n l + l k n)$ and a projection cost of $O(n l \log l)$. The update of $\mathbf{A}_v$ involves an SVD on a $d_v \times l$ matrix, yielding $O(d_v l^2)$, while updating $\mathbf{H}_v$ costs $O(n l k)$. The update of $\mathbf{Y}$ involves forming $\sum_v \mathbf{H}_v^T \mathbf{Z}_v$ with cost $O(\sum_v n l k)$ and a small-scale SVD of size $k \times k$, which is negligible since $k \ll n$. Updating α is $O(V)$ and does not affect the overall complexity. Therefore, letting $d_v, l, k \ll n$, the per-iteration complexity of the proposed framework is $O\left(\sum_{v=1}^V (d_v n l + d_v l^2 + n l k + n l \log l)\right)$, which is dominated by $O(\sum_{v=1}^V d_v n l)$ in practice. As a result, FPMCAA scales approximately linearly with the number of samples and is suitable for large-scale multi-view clustering.

2) *Space Complexity*: In terms of memory consumption, the main storage requirements come from the data matrices $\mathbf{X}_v \in \mathbb{R}^{d_v \times n}$ for each view, the anchor bases $\mathbf{A}_v \in \mathbb{R}^{d_v \times l}$, and the anchor representations $\mathbf{Z}_v \in \mathbb{R}^{l \times n}$. Thus, the space complexity for each view is $O(d_v n + d_v l + l n)$. In addition, the alignment matrices $\mathbf{H}_v \in \mathbb{R}^{l \times k}$ and the consensus indicator $\mathbf{Y} \in \mathbb{R}^{n \times k}$ contribute $O(l k + n k)$. Therefore, the overall space complexity is $O(\sum_{v=1}^V (d_v n + d_v l + l n + l k) + n k)$. Since $l, k \ll n$, the dominant term is $O(\sum_{v=1}^V d_v n)$, which grows linearly with the dataset size.

D. Convergence Analysis

To further ensure the theoretical soundness of FPMCAA framework, we provide an analysis of its convergence behavior. Since the objective function involves multiple variables updated in an alternating manner, it is crucial to establish that the optimization process yields a monotonically decreasing sequence of objective values and eventually converges to a stationary solution. Let $\Theta = \mathbf{A}_v, \mathbf{Z}_v, \mathbf{H}_v, \mathbf{Y}, \alpha$ denote the collection of all optimization variables. The overall objective function can be formulated as

$$\mathcal{J}(\Theta) = \sum_{v=1}^V (\alpha_v^2 \|\mathbf{X}_v - \mathbf{A}_v \mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v\|_F^2 + \|\mathbf{Z}_v - \mathbf{H}_v \mathbf{Y}\|_F^2), \quad (29)$$

which is clearly lower bounded by 0 since each term is non-negative.

For the update of $\mathbf{A}_v$, when other variables are fixed, the subproblem with respect to $\mathbf{A}_v$ admits a closed-form solution under the orthogonality constraint. Therefore, the following inequality holds:

$$\begin{aligned} \mathcal{J}(\mathbf{A}_v^t, \mathbf{Z}_v^t, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t) \\ \geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^t, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t). \end{aligned} \quad (30)$$

For the update of $\mathbf{Z}_v$, with $\mathbf{A}_v$, $\mathbf{H}_v$, $\mathbf{Y}$, and α fixed, solving the quadratic subproblem decreases the objective value:

$$\begin{aligned} \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^t, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t) \\ \geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t). \end{aligned} \quad (31)$$

For the update of $\mathbf{H}_v$, the subproblem also admits an exact solution, yielding

$$\begin{aligned} \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t) \\ \geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^t, \alpha^t). \end{aligned} \quad (32)$$

For the update of $\mathbf{Y}$, optimization under the orthogonality constraint can be solved via eigen-decomposition, which ensures

$$\begin{aligned} \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^t, \alpha^t) \\ \geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^{t+1}, \alpha^t). \end{aligned} \quad (33)$$

Finally, for the update of α , the closed-form solution under simplex constraints leads to

$$\begin{aligned} \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^{t+1}, \alpha^t) \\ \geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^{t+1}, \alpha^{t+1}). \end{aligned} \quad (34)$$

By combining the above results, we obtain the inequality chain:

$$\begin{aligned} \mathcal{J}(\Theta^t) &\geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^t, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t) \\ &\geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^t, \mathbf{Y}^t, \alpha^t) \\ &\geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^t, \alpha^t) \\ &\geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^{t+1}, \alpha^t) \\ &\geq \mathcal{J}(\mathbf{A}_v^{t+1}, \mathbf{Z}_v^{t+1}, \mathbf{H}_v^{t+1}, \mathbf{Y}^{t+1}, \alpha^{t+1}) \\ &= \mathcal{J}(\Theta^{t+1}). \end{aligned} \quad (35)$$

Hence, the sequence $\mathcal{J}(\Theta^t)$ is monotonically non-increasing and bounded below by zero, which guarantees convergence. Moreover, since each subproblem admits an exact or closed-form solution under the given constraints, any limit point of the sequence corresponds to a stationary point of the original optimization problem. Therefore, the proposed FPMCAA method is guaranteed to converge.

IV. EXPERIMENTS

This section provides a comprehensive evaluation of FPMCAA through clustering accuracy comparison, ablation experiments, convergence verification, and runtime analysis.

A. Experimental Setting

We evaluate clustering performance using six standard external metrics: Accuracy (ACC), Normalized Mutual Information (NMI), Purity, F-score, Adjusted Rand Index (ARI), and Precision, where higher scores indicate better clustering quality. To ensure reliable results, the k -means step is repeated 50 times under different random seeds, and the mean and standard deviation are reported. All experiments are conducted on Ubuntu

TABLE II
DETAILS OF THE EXPERIMENTAL MULTI-VIEW DATASETS

Datasets	Samples	Clusters	Views	Features
ORL	400	40	4	512, 59, 864, 254
BDGP	2500	5	2	1750, 79
Cifar-100	50000	100	3	512, 2048, 1024
MNIST	60000	10	3	342, 1024, 64
YTF-10	38654	10	4	944, 576, 512, 640
YTF-20	63896	20	4	944, 576, 512, 640
YTF-50	126054	50	4	944, 576, 512, 640
YTF-100	195537	100	4	944, 576, 512, 640
YTF-200	286006	200	4	944, 576, 512, 640

with MATLAB R2020a, running on a 2.4 GHz Intel CPU and 64 GB of RAM.

B. Datasets and Compared Methods

1) *Multi-View Datasets*: To thoroughly evaluate the effectiveness and scalability of the proposed method, we conduct experiments on nine widely-used multi-view benchmark datasets, covering a broad range of domains, sample sizes, and feature types. Detailed descriptions are as follows:

- 1) *ORL*:¹ A face image dataset containing 400 samples from 40 individuals. Each sample is represented by four feature views: intensity (512), LBP (59), Gabor (864), and shape descriptors (254).
- 2) *BDGP*:² A biological image dataset with 2,500 samples from 5 classes. Each sample is described by two views of dimensions 1750 and 79.
- 3) *Cifar-100*:³ An image dataset with 50,000 samples from 100 object categories. Each sample is represented by three deep feature views of dimensions 512, 2048, and 1024.
- 4) *MNIST*:⁴ A handwritten digit dataset with 60,000 samples from 10 categories. Each image is represented by three feature views with dimensions 342, 1024, and 64.
- 5) *YouTube Face (YTF)*:⁵ A large-scale face dataset. Five subsets with increasing cluster numbers are used: YTF-10, YTF-20, YTF-50, YTF-100, and YTF-200. Each sample has four feature views with dimensions 944, 576, 512, and 640.

These datasets span applications such as face recognition, biological imaging, object classification, and handwriting analysis. They also vary markedly in scale, from a 400 to over 286,000 samples, and in the number of views and feature dimensionality, forming a comprehensive testbed for performance and scalability evaluation, as shown in Table II.

2) *Compared Methods*: To demonstrate the superiority of our method, we selected 14 state-of-the-art multi-view clustering methods for comparison, as detailed below.

- 1) *OMSC* [41] jointly learns anchors and affinity graphs in a unified framework to improve clustering performance and scalability.

¹<https://www.cl.cam.ac.uk/research/dtg/attarchive/facedatabase.html>

²<https://www.fruitfly.org/>

³<http://www.cs.toronto.edu/kriz/cifar.html>

⁴<http://yann.lecun.com/exdb/mnist/>

⁵<https://www.micc.unifi.it/resources/datasets/e-ytf/>

- 2) FPMVS-CAG [37] integrates anchor selection and sub-space graph construction into a single model to avoid heuristic sampling and enhance efficiency.
- 3) EOMSC-CA [45] combines anchor learning and graph construction with a consensus mechanism to reduce complexity and improve clustering quality.
- 4) EEOMVC [47] builds latent partition representations from anchor graphs and fuses them for direct clustering without heavy processing.
- 5) FLMSC [48] factorizes view-specific representations into compact components to reduce SVD costs while preserving structural consistency.
- 6) FSMSC [49] uses a shared-anchor learning scheme with guided feature selection to balance consistency and view diversity.
- 7) OSMvSC [50] unifies dictionary learning and anchor representation to obtain clustering assignments with linear complexity.
- 8) OMVCDR [51] projects each view into diverse latent spaces and fuses them with auto-weighting for unified clustering.
- 9) AGLSR [20] learns adaptive affinity graphs per view and aligns spectral embeddings within a unified optimization framework.
- 10) OMVFC [33] constructs cross-view anchor graphs and jointly optimizes membership matrices to achieve one-step fuzzy clustering.
- 11) MvSST [38] builds view-specific bipartite graphs and fuses them via a self-weighted strategy to generate cluster labels directly.
- 12) NpGC [36] adopts view-related and shared anchors with a consensus bipartite graph to capture complementarities without hyper-parameters.
- 13) SMVAGC-SF [44] jointly optimizes anchor graph construction and structural alignment with flexible fusion across views.
- 14) ESTMC [42] combines anchor representations with tensor-based low-rank learning to handle large-scale multi-view data efficiently.

For all baselines, we follow the parameter settings reported in the original papers and fine-tune them within the recommended ranges to achieve their best performance. For completeness, Table III summarizes the main computational complexities, the number of tunable parameters, and the maximum dataset sizes reported for all compared methods.

C. Clustering Performance Comparison

We evaluate the proposed FPMCAA model on nine multi-view datasets, with sample sizes ranging from 400 to over 286,006, to comprehensively verify its clustering performance and effectiveness. The detailed results are summarized in Table IV, where the best and second-best results are highlighted in red and blue, respectively. Based on the results, the following conclusions can be drawn:

- 1) FPMCAA achieves stable superiority from hundred-scale (e.g., ORL, BDGP) to hundred-thousand-scale (e.g.,

TABLE III
DETAILS OF OUR METHOD AND THE COMPARED METHODS

Method	Number of Parameters	Time Complexity	Max Reported
OMSC	1	$O(nl^3 + nlk)$	195,537
FPMVS	0	$O(nl)$	101,499
EOMSC	0	$O(nl^2)$	101,499
EEOMVC	2	$O(nl^2 + nk)$	10,800
FLMSC	2	$O(d_v n^2)$	10,800
FSMSC	3	$O(nk^3 + nk^2)$	60,000
OSMvSC	2	$O(nkd)$	30,000
OMVCDR	1	$O(ndl)$	63,896
AGLSR	2	$O(n^2 dl)$	4,485
OMVFC	2	$O(nl)$	30,000
MvSST	0	$O(nl + l^3)$	10,000
NpGC	0	$O(ndk + nk^3)$	126,054
SMVAGC	1	$O(ndk)$	101,499
ESTMC	1	$O(nl \log n)$	50,000
Our	0	$O(\sum_{v=1}^V d_v nl)$	286,006

CIFAR, YTF-200). Regardless of view heterogeneity or sample size, it consistently delivers the best or near-best performance, confirming its robustness, scalability, and applicability to other scenarios.

- 2) Compared with shared-anchor methods such as OMSC, FPMVS, EOMSC, and FSMSC, which can introduce bias due to view heterogeneity, our model allows anchors to be view-specific and jointly optimized, leading to better adaptability and representation. This flexibility helps better capture heterogeneity in different views. On YTF-10 and MNIST, FPMCAA exceeds FSMSC by 9.41% and 9.58% ACC respectively, validating the benefits of flexible anchors.
- 3) Unlike methods without graph alignment mechanisms (e.g., OSMvSC and OMVCDR), FPMCAA aligns view-specific anchor graphs during optimization, promoting cross-view structural consistency. For instance, on Cifar-100, our method improves ACC by 10.45% over OMVCDR, highlighting the importance of dynamically aligning anchor graphs structures during learning.
- 4) On large-scale datasets such as YTF-100 and YTF-200, only a few baselines (e.g., OMSC, FPMVS, EOMSC, NpGC, ESTMC) can run successfully. Among them, FPMCAA achieves the highest clustering quality, thanks to its flexible anchor construction and alignment-based fusion, which effectively capture cross-view structural information.
- 5) Although FLMSC and AGLSR capture inter-sample structural information via full-sample affinity matrices at quadratic cost, their clustering performance still falls slightly short of that achieved by the linear-complexity FPMCAA. Our model achieves better or comparable results while maintaining lower computational costs, demonstrating the strength of its anchor-based design. Additionally, even compared with models that also claim linear complexity, such as NpGC, SMVAGC-SF, and

TABLE IV
CLUSTERING RESULTS (MEAN % $\pm$ STD %) ON NINE BENCHMARK DATASETS. '-' MEANS UNABLE TO RUN DUE TO MEMORY LIMITATION

Methods	OMSC	FPMVS	EOMSC	EEOMVC	FLMSC	FSMSC	OSMvSC	OMVCDR	AGLSR	OMVFC	MvSST	NpGC	SMVAGC-SF	ESTMC	Ours
ACC															
ORL	57.25	53.82 $\pm$ 2.06	62.25	76.00	39.54	72.15 $\pm$ 3.71	58.50	76.00	77.50	73.50 $\pm$ 0.95	68.00	72.77 $\pm$ 3.72	52.15 $\pm$ 0.03	77.50	79.90 $\pm$ 1.41
BDGP	60.08	55.60 $\pm$ 3.74	92.04	77.68	58.76	63.80 $\pm$ 5.92	70.72	48.32	76.64	60.52 $\pm$ 2.80	38.96	48.30 $\pm$ 0.10	85.61 $\pm$ 10.67	82.08	91.72 $\pm$ 0.00
Cifar-100	87.01	61.31 $\pm$ 2.95	91.44	-	-	84.34 $\pm$ 2.94	84.77	84.06	-	93.24 $\pm$ 1.53	-	86.47 $\pm$ 2.79	87.85 $\pm$ 2.35	93.49	94.51 $\pm$ 1.06
MNIST	89.24	82.87 $\pm$ 6.68	91.57	-	-	89.27 $\pm$ 6.09	85.70	91.74	-	90.43 $\pm$ 2.69	-	88.77 $\pm$ 8.48	87.72 $\pm$ 5.40	98.44	98.85 $\pm$ 0.00
YTF-10	78.26	67.51 $\pm$ 3.61	74.43	81.19	-	70.14 $\pm$ 7.46	79.86	73.82	-	62.29 $\pm$ 0.91	-	72.16 $\pm$ 6.95	77.43 $\pm$ 5.98	77.47	79.55 $\pm$ 1.24
YTF-20	74.46	62.29 $\pm$ 3.31	73.82	71.29	-	69.04 $\pm$ 5.09	69.47	50.44	-	-	-	65.76 $\pm$ 5.13	69.62 $\pm$ 0.05	75.46	76.51 $\pm$ 1.97
YTF-50	70.65	63.68 $\pm$ 2.47	74.22	-	-	-	-	-	-	-	-	67.47 $\pm$ 2.57	-	74.13	74.81 $\pm$ 1.54
YTF-100	63.85	49.11 $\pm$ 1.80	68.86	-	-	-	-	-	-	-	-	63.69 $\pm$ 2.37	-	65.64	69.58 $\pm$ 1.27
YTF-200	62.34	-	-	-	-	-	-	-	-	-	-	59.63 $\pm$ 1.41	-	-	62.81 $\pm$ 1.11
NMI															
ORL	79.52	77.43 $\pm$ 1.03	81.22	86.41	36.12	87.72 $\pm$ 1.66	67.54	86.41	75.03	85.41 $\pm$ 0.22	83.85	88.13 $\pm$ 1.72	73.31 $\pm$ 0.01	88.82	77.18 $\pm$ 0.02
BDGP	40.76	37.56 $\pm$ 4.18	82.89	64.01	49.78	41.12 $\pm$ 3.92	55.46	33.19	70.84	31.19 $\pm$ 0.60	39.43	22.36 $\pm$ 0.16	83.68 $\pm$ 5.71	64.33	90.10 $\pm$ 0.39
Cifar-100	97.92	87.84 $\pm$ 1.73	98.54	-	-	97.31 $\pm$ 0.51	97.49	94.93	-	98.24 $\pm$ 1.18	-	97.77 $\pm$ 0.46	97.90 $\pm$ 0.40	98.92	99.24 $\pm$ 0.15
MNIST	93.41	91.59 $\pm$ 2.75	90.68	-	-	93.84 $\pm$ 2.13	90.28	89.01	-	83.64 $\pm$ 4.35	-	92.91 $\pm$ 3.00	87.14 $\pm$ 2.03	95.73	96.04 $\pm$ 0.00
YTF-10	82.83	73.38 $\pm$ 2.52	81.12	79.75	-	76.64 $\pm$ 3.57	76.08	75.11	-	64.39 $\pm$ 1.08	-	79.59 $\pm$ 3.15	79.84 $\pm$ 3.30	76.94	83.27 $\pm$ 0.48
YTF-20	81.70	74.90 $\pm$ 1.90	80.81	78.06	-	78.61 $\pm$ 2.05	76.53	69.32	-	-	-	77.09 $\pm$ 1.90	79.47 $\pm$ 0.02	80.45	80.76 $\pm$ 0.94
YTF-50	86.00	81.55 $\pm$ 1.16	85.00	-	-	-	-	-	-	-	-	83.69 $\pm$ 0.86	-	85.14	85.15 $\pm$ 0.47
YTF-100	81.55	72.74 $\pm$ 1.38	82.86	-	-	-	-	-	-	-	-	83.16 $\pm$ 0.77	-	82.77	84.31 $\pm$ 0.43
YTF-200	80.58	-	-	-	-	-	-	-	-	-	-	83.18 $\pm$ 0.47	-	-	83.54 $\pm$ 0.31
Purity															
ORL	60.50	58.42 $\pm$ 1.97	68.00	77.50	40.00	75.85 $\pm$ 3.16	65.50	77.50	79.00	74.45 $\pm$ 1.66	71.50	76.51 $\pm$ 3.33	55.19 $\pm$ 0.02	79.75	81.30 $\pm$ 0.75
BDGP	60.48	58.96 $\pm$ 4.31	92.04	77.68	64.68	64.39 $\pm$ 4.88	70.72	51.68	76.64	60.52 $\pm$ 2.63	38.96	48.94 $\pm$ 0.10	87.01 $\pm$ 9.01	82.08	90.84 $\pm$ 0.00
Cifar-100	87.02	61.9 $\pm$ 2.90	91.51	-	-	88.14 $\pm$ 2.19	95.74	87.93	-	93.50 $\pm$ 1.41	-	90.34 $\pm$ 2.00	91.23 $\pm$ 1.68	94.98	96.78 $\pm$ 0.60
MNIST	89.24	84.88 $\pm$ 6.68	91.61	-	-	89.34 $\pm$ 6.05	95.19	91.74	-	90.43 $\pm$ 4.38	-	91.69 $\pm$ 6.07	88.70 $\pm$ 4.69	98.44	98.60 $\pm$ 0.00
YTF-10	83.03	70.09 $\pm$ 3.94	78.13	80.19	-	76.02 $\pm$ 4.80	79.86	74.22	-	67.24 $\pm$ 1.15	-	79.74 $\pm$ 3.59	79.73 $\pm$ 5.35	78.04	81.12 $\pm$ 0.61
YTF-20	77.31	66.25 $\pm$ 3.60	79.87	74.13	-	75.30 $\pm$ 3.74	76.21	63.82	-	-	-	72.53 $\pm$ 3.43	75.88 $\pm$ 0.03	79.34	79.52 $\pm$ 1.62
YTF-50	76.86	65.93 $\pm$ 2.67	77.09	-	-	-	-	-	-	-	-	74.14 $\pm$ 1.93	-	79.55	80.27 $\pm$ 0.99
YTF-100	68.02	50.33 $\pm$ 1.96	73.15	-	-	-	-	-	-	-	-	72.20 $\pm$ 1.48	-	74.11	76.26 $\pm$ 0.79
YTF-200	66.35	-	-	-	-	-	-	-	-	-	-	70.69 $\pm$ 0.93	-	-	72.82 $\pm$ 0.66
Fscore															
ORL	42.82	43.64 $\pm$ 2.90	63.28	67.90	50.90	65.53 $\pm$ 3.94	44.70	67.90	73.03	57.97 $\pm$ 1.21	51.14	66.59 $\pm$ 3.95	37.95 $\pm$ 0.02	70.08	72.50 $\pm$ 0.96
BDGP	47.99	46.85 $\pm$ 2.71	85.07	69.76	56.90	50.64 $\pm$ 3.12	51.64	38.96	74.08	42.54 $\pm$ 1.84	43.66	34.93 $\pm$ 0.10	82.08 $\pm$ 8.30	67.23	82.39 $\pm$ 0.00
Cifar-100	88.40	59.72 $\pm$ 6.31	91.76	-	-	86.11 $\pm$ 2.64	86.79	84.59	-	92.59 $\pm$ 2.04	-	88.48 $\pm$ 2.33	89.35 $\pm$ 2.03	94.25	95.84 $\pm$ 1.27
MNIST	89.25	84.78 $\pm$ 6.80	88.95	-	-	89.39 $\pm$ 5.29	86.06	90.60	-	83.46 $\pm$ 4.73	-	89.53 $\pm$ 6.74	84.01 $\pm$ 4.47	96.92	97.23 $\pm$ 0.00
YTF-10	74.63	63.20 $\pm$ 4.10	68.94	70.56	-	66.18 $\pm$ 6.39	67.67	66.79	-	58.35 $\pm$ 0.89	-	70.44 $\pm$ 6.30	72.75 $\pm$ 4.27	69.73	76.49 $\pm$ 0.72
YTF-20	68.35	57.02 $\pm$ 4.11	67.96	65.89	-	63.35 $\pm$ 5.14	61.09	48.7	-	-	-	60.64 $\pm$ 4.68	64.68 $\pm$ 0.05	67.99	68.78 $\pm$ 1.77
YTF-50	64.15	58.08 $\pm$ 3.58	64.63	-	-	-	-	-	-	-	-	61.30 $\pm$ 3.11	-	65.15	65.76 $\pm$ 1.90
YTF-100	52.90	26.63 $\pm$ 4.38	57.85	-	-	-	-	-	-	-	-	53.64 $\pm$ 2.82	-	51.81	58.28 $\pm$ 2.06
YTF-200	40.75	-	-	-	-	-	-	-	-	-	-	46.29 $\pm$ 2.32	-	-	49.12 $\pm$ 2.10
ARI															
ORL	32.05	36.11 $\pm$ 3.69	52.80	67.15	30.47	60.74 $\pm$ 4.23	43.22	67.15	71.35	56.80 $\pm$ 1.28	35.80	61.65 $\pm$ 4.03	30.89 $\pm$ 0.03	69.34	71.83 $\pm$ 0.98
BDGP	45.01	46.06 $\pm$ 4.91	84.53	61.84	54.26	49.05 $\pm$ 5.15	38.23	19.89	66.58	26.82 $\pm$ 3.10	18.46	34.38 $\pm$ 0.10	77.50 $\pm$ 12.71	58.47	77.93 $\pm$ 0.00
Cifar-100	79.32	47.72 $\pm$ 4.46	85.28	-	-	78.64 $\pm$ 3.81	86.65	84.43	-	87.79 $\pm$ 2.73	-	82.46 $\pm$ 3.41	83.67 $\pm$ 3.04	93.19	93.37 $\pm$ 2.20
MNIST	82.14	75.45 $\pm$ 10.17	85.00	-	-	82.55 $\pm$ 8.94	84.38	90.11	-	81.60 $\pm$ 5.27	-	85.63 $\pm$ 9.81	80.36 $\pm$ 7.13	96.58	96.92 $\pm$ 0.00
YTF-10	72.69	58.78 $\pm$ 7.11	69.51	67.40	-	64.73 $\pm$ 7.57	63.44	62.72	-	52.45 $\pm$ 1.18	-	68.74 $\pm$ 7.98	73.43 $\pm$ 4.21	66.05	73.31 $\pm$ 0.81
YTF-20	62.53	50.34 $\pm$ 6.01	66.18	63.82	-	61.31 $\pm$ 6.20	58.64	45.21	-	-	-	58.22 $\pm$ 5.64	62.76 $\pm$ 0.05	68.21	66.91 $\pm$ 1.89
YTF-50	64.01	48.83 $\pm$ 4.89	59.82	-	-	-	-	-	-	-	-	57.02 $\pm$ 3.93	-	64.31	64.94 $\pm$ 2.00
YTF-100	43.76	16.67 $\pm$ 3.35	50.99	-	-	-	-	-	-	-	-	49.37 $\pm$ 3.81	-	51.17	57.75 $\pm$ 2.09
YTF-200	29.63	-	-	-	-	-	-	-	-	-	-	41.06 $\pm$ 3.19	-	-	48.75 $\pm$ 2.12
Precision															
ORL	64.50	55.45 $\pm$ 0.75	63.61	67.52	45.38	67.19 $\pm$ 3.92	52.56	67.52	66.13	48.08 $\pm$ 2.76	68.50	62.45 $\pm$ 4.15	49.39 $\pm$ 0.02	65.15	68.85 $\pm$ 1.17
BDGP	51.39	47.67 $\pm$ 2.95	75.61	67.11	59.80	52.64 $\pm$ 1.67	47.69	32.68	65.92	39.54 $\pm$ 0.92	28.21	35.51 $\pm$ 0.11	79.18 $\pm$ 2.89	63.66	81.45 $\pm$ 0.00
Cifar-100	90.82	76.96 $\pm$ 0.23	90.59	-	-	95.22 $\pm$ 0.98	95.12	79.11	-	90.99 $\pm$ 1.18	-	92.52 $\pm$ 0.92	95.90 $\pm$ 0.81	90.71	98.48 $\pm$ 0.24
MNIST	90.71	90.78 $\pm$ 0.09	94.22	-	-	97.07 $\pm$ 0.10	92.28	95.54	-	92.27 $\pm$ 4.88	-	94.15 $\pm$ 3.03	88.27 $\pm$ 1.34	96.87	97.19 $\pm$ 0.00
YTF-10	72.69	68.95 $\pm$ 1.86	68.37	74.04	-	67.85 $\pm$ 5.48	71.87	66.86	-	51.38 $\pm$ 0.68	-	72.45 $\pm$ 5.07	76.20 $\pm$ 3.05	70.18	77.13 $\pm$ 0.79
YTF-20	65.37	66.35 $\pm$ 1.26	64.83	68.09	-	65.64 $\pm$ 4.28	65.33	42.41	-	-	-	63.37 $\pm$ 3.87	66.81 $\pm$ 0.04	66.18	67.90 $\pm$ 2.24
YTF-50	61.61	62.09 $\pm$ 0.33	60.28	-	-	-	-	-	-	-	-	66.39 $\pm$ 2.62	-	61.94	63.12 $\pm$ 2.57
YTF-100	54.88	55.41 $\pm$ 0.8	53.86	-	-	-	-	-	-	-	-	58.84 $\pm$ 1.94	-	47.59	56.04 $\pm$ 3.05
YTF-200	33.22	-	-	-	-	-	-	-	-	-	-	43.19 $\pm$ 1.62	-	-	44.97 $\pm$ 3.12

ESTMC, our method still outperforms them in both clustering performance. For example, on YTF-50, FPMCAA achieves 74.81% ACC, outperforming ESTMC and NpGC by 0.68% and 7.34%, respectively, thereby corroborating both the efficiency and efficacy of FPMCAA.

- 6) While FPMVS, MvSST, and EOMSC are also parameter-free, they either depend on fixed anchor graph or shared anchors. In contrast, FPMCAA introduces adaptive anchor construction and alignment-based fusion, enabling superior performance. On YTF-100, it surpasses FPMVS and NpGC by 20.47% and 5.89% in ACC, respectively.
- 7) The proposed FPMCAA also demonstrates high stability. On datasets such as ORL, Cifar-100, and YTF-10, the standard deviation of its metrics is consistently small, indicating robustness to different initializations and data

variations. This can be attributed to its joint optimization of anchors, graphs, and latent embeddings.

D. Sensitivity to the Number of Anchors

Although our framework does not require parameter tuning during training, the number of anchors is initialized according to the number of clusters k , which is a common convention in anchor-based clustering. To evaluate influence, we vary the anchor size from k to $7k$ and report accuracy and runtime on nine datasets in Fig. 2.

The results show that increasing the number of anchors leads to only minor performance fluctuations. Across all datasets, the variation is small, indicating that using k anchors without additional tuning is sufficient to provide stable and

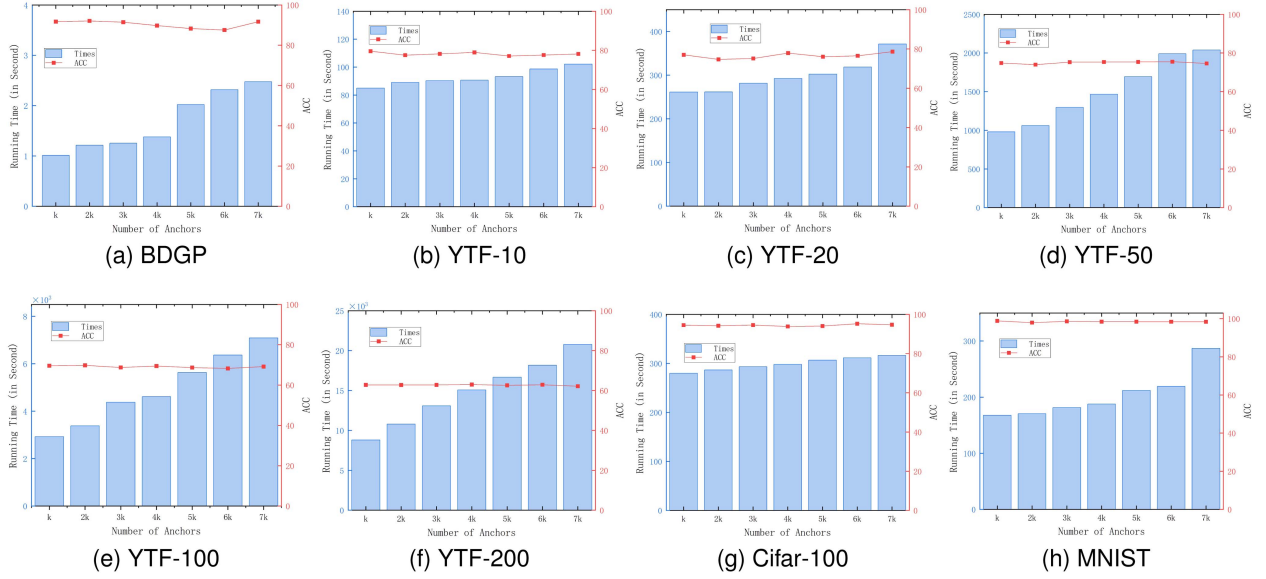


Fig. 2. Left-axis bars: runtime (seconds); right-axis line: ACC (%). Both vary with the number of anchors on the x-axis. Bar height reflects execution time, while markers on the line indicate clustering accuracy.

TABLE V
ABLATION STUDY ON NINE LARGE-SCALE DATASETS (ACCURACY %)

Datasets	ORL	BDGP	YTF-10	YTF-20	YTF-50	YTF-100	YTF-200	Cifar-100	MNIST
FPMCAA- α	79.08 $\pm$ 1.97	88.68 $\pm$ 0.03	79.32 $\pm$ 2.29	75.58 $\pm$ 1.92	73.16 $\pm$ 1.89	65.45 $\pm$ 1.42	62.80 $\pm$ 0.82	94.19 $\pm$ 1.00	98.49 $\pm$ 0.00
FPMCAA-A	75.07 $\pm$ 2.56	88.24 $\pm$ 0.00	71.49 $\pm$ 0.00	65.29 $\pm$ 0.00	69.97 $\pm$ 0.00	63.45 $\pm$ 0.00	62.08 $\pm$ 0.00	86.06 $\pm$ 0.00	98.48 $\pm$ 0.00
FPMCAA-Z	55.73 $\pm$ 2.52	69.32 $\pm$ 0.00	71.55 $\pm$ 1.67	72.14 $\pm$ 2.18	71.07 $\pm$ 2.68	64.27 $\pm$ 1.72	59.43 $\pm$ 1.67	91.37 $\pm$ 1.07	92.49 $\pm$ 0.00
FPMCAA	79.90 $\pm$ 1.41	91.72 $\pm$ 0.00	79.55 $\pm$ 1.24	76.51 $\pm$ 1.97	74.81 $\pm$ 1.54	69.26 $\pm$ 1.27	62.88 $\pm$ 1.11	95.49 $\pm$ 0.77	98.85 $\pm$ 0.00

competitive performance. In contrast, the computational cost increases noticeably as the anchor size grows. Larger anchor sets introduce additional overhead in reconstruction and graph alignment, which significantly increases the running time, particularly on large-scale datasets such as Cifar-100, YTF-50, and YTF-100. Therefore, selecting k anchors as the default setting achieves a favorable balance between accuracy and efficiency. These findings demonstrate that the proposed method is robust to variations in anchor size and does not rely on hyperparameter tuning.

E. Ablation Study

To evaluate the contribution of each major component in our framework, we design three simplified variants for comparison: **FPMCAA- α** removes the adaptive view-weight parameter α and assumes that all views contribute equally; **FPMCAA-A** replaces view-specific anchor learning with a fixed anchor set shared across all views; **FPMCAA-Z** discards the graph alignment module and directly performs spectral clustering on the averaged anchor graph $\mathbf{Z}_{\text{fusion}} = \frac{1}{V} \sum_{v=1}^V \mathbf{Z}_v$.

The experimental results are reported in Table V. FPMCAA- α shows only a slight drop in performance, indicating that adaptive view weighting provides additional gains but is not the dominant factor. In contrast, once view-specific anchors are removed in FPMCAA-A or graph alignment is discarded

in FPMCAA-Z, the performance drops more clearly. The decline is especially significant on datasets such as Cifar-100 and YTF-200 where sample size is large or view heterogeneity is strong. These results confirm the importance of learning anchors tailored to each view and aligning them before fusion so that complementary structural information can be fully exploited.

Overall, each module contributes positively to the final performance. The complete version of FPMCAA achieves the best results on all datasets, validating both the necessity and cooperative effect of its core components.

F. Visualization Analysis

To qualitatively evaluate the effect of cross-view alignment, we visualize the learned representations using t-SNE on the BDGP and YTF-10 datasets. We compare three cases: the original features, the representations obtained without the alignment term, and those produced by the full FPMCAA model. As illustrated in Fig. 4, the original features exhibit significant overlap among different classes, while removing the alignment term results in partially improved but still mixed clusters. In contrast, the full model produces more compact and clearly separated clusters, providing qualitative evidence that the proposed cross-view alignment enhances the discriminative structure of the learned representation.

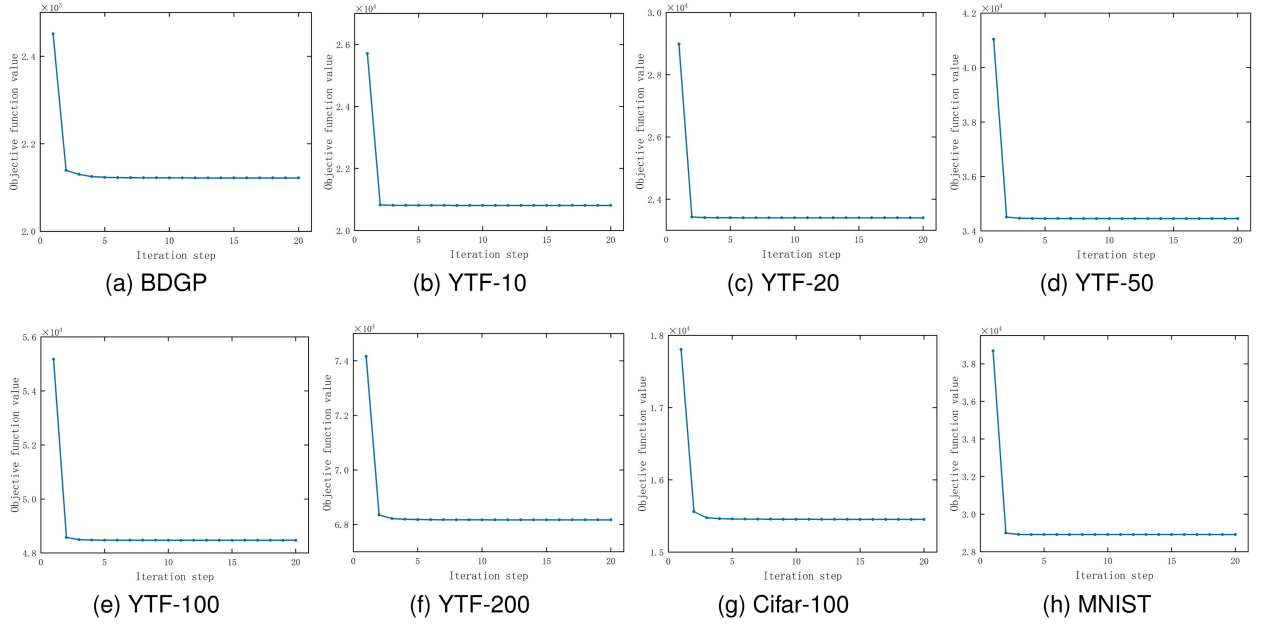


Fig. 3. Convergence curves of the proposed Algorithm 1.

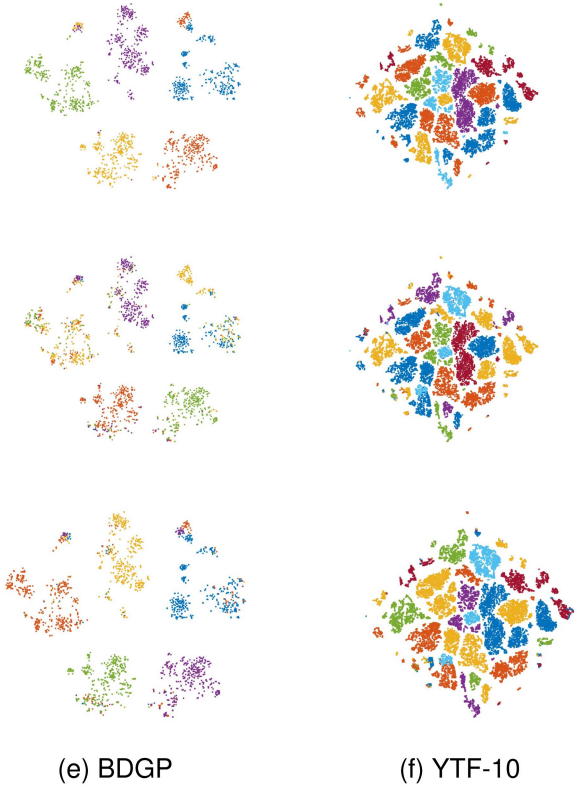


Fig. 4. t-SNE visualization on BDGP and YTF-10 datasets. For each dataset, three rows illustrate: (top) raw features, (middle) representations without the alignment term (FPMCAA-Z), and (bottom) the full FPMCAA model.

G. Running Time Comparison

To evaluate time efficiency, we conduct experiments on both small-scale datasets (e.g., ORL and BDGP) and large-scale

datasets (e.g., Cifar-100, MNIST, and the YTF series). The complete running time is summarized in Table VI.

Overall, FPMCAA demonstrates consistent and favorable efficiency across datasets of different sizes. As the dataset size increases, several existing approaches show rapidly increasing runtime or become inapplicable, whereas FPMCAA remains both executable. For example, on Cifar-100 and MNIST, the running times are 334.69 s and 167.74 s, respectively, showing a clear advantage over other baselines that are still runnable. A similar trend is observed on the YTF datasets: on YTF-10 and YTF-20, the runtime stays within a few hundred seconds, and even on YTF-200, our method completes clustering successfully while many competing models fail to run.

H. Convergence Study

To empirically validate the convergence behavior of the proposed FPMCAA framework, we record the objective value at each iteration on eight representative datasets, including both small-scale (e.g., BDGP) and large-scale benchmarks (e.g., Cifar-100, YTF-200).

As illustrated in Fig. 3, all curves exhibit a monotonic decreasing trend, confirming the descent of the objective function during optimization. Specifically, the objective value shows a steep decline in the early iterations, and then quickly stabilizes to a plateau. This rapid early decrease reflects the efficiency of the update rules for anchor learning, graph construction, and consensus alignment. On most datasets, convergence is achieved within fewer than 10 iterations, and all curves flatten before the 20 iteration, demonstrating both rapid convergence and numerical stability. Moreover, the absence of oscillations across different datasets indicates that the alternating optimization strategy is robust to variations in data scale, feature dimensionality, and view heterogeneity. The stable behavior on larger datasets such as

TABLE VI
RUNNING TIME (IN SECONDS) OF DIFFERENT METHODS ON DATASETS. ‘-’ MEANS UNABLE TO RUN DUE TO MEMORY LIMITATION

Datasets	OMSC	FPMVS	EOMSC	EEOMVC	FSMSC	OSMvSC	OMVCDR	OMVFC	MvSST	NpGC	SMVAGC-SF	ESTMC	Ours
ORL	2.22	3.38	8.72	1.67	1.99	1.21	4.28	1.22	7.35	4.81	4.79	1.02	0.79
BDGP	12.94	11.03	9.13	2.76	12.70	2.59	11.13	3.78	26.96	5.24	9.77	4.09	2.82
Cifar-100	4524.71	1016.80	423.62	708.67	4913.00	7572.90	1731.00	648.78	3690.43	3761.90	3832.40	530.91	334.69
MNIST	4460.94	689.99	255.60	347.06	6255.00	5997.20	1757.40	355.79	9811.82	1554.10	2109.20	217.12	167.74
YTF-10	1657.82	778.38	159.39	184.18	2246.30	442.03	1396.50	71.67	-	555.23	879.54	147.38	84.93
YTF-20	3624.96	1638.40	346.49	350.90	4716.50	720.81	3754.90	-	-	1274.30	2315.20	560.50	261.42
YTF-50	8497.85	4513.70	1847.20	861.09	-	-	-	-	-	4784.70	-	2511.75	979.98
YTF-100	16002.42	9434.90	2566.40	-	-	-	-	-	-	16738.00	-	4157.30	2923.39
YTF-200	90078.63	-	-	-	-	-	-	-	-	122620.00	-	-	8795.25

YTF-100 and YTF-200 further verifies that FPMCAA remains well-behaved under large-sample settings.

These empirical results confirm that FPMCAA converges reliably in practice and requires only a small number of iterations before stabilization.

V. CONCLUSION

In this paper, we propose a unified and parameter-free multi-view clustering framework, which jointly optimizes view-specific anchor construction and consensus assignment without requiring any manually tuned hyper-parameters. By introducing an adaptive anchor learning strategy and a unified alignment mechanism, our method effectively captures both local structures and cross-view consistency. Extensive experiments on multiple benchmark datasets demonstrate that the proposed framework achieves competitive or superior clustering performance compared to state-of-the-art methods, while maintaining linear complexity with respect to the number of samples. In the future, we plan to extend the model to address scenarios with incomplete views, and explore integrating semantic priors to further enhance representation quality.

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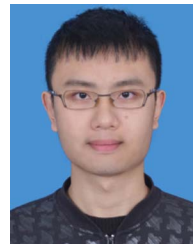


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